Natural mortality-at-age vectors from the hake predation model to be used in the Reference Set Operating models for the OMP-2022 revision

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Summary

The 2021 Reference Case (RC) model uses mortality-at-age vectors from the 2018 version of the hake predation model (Ross-Gillespie and Butterworth, 2018). This predation model has since been updated, and corresponding updated mortality-at-age vectors needed to be used in the Operating Models for the 2022 OMP revision. Use of these updated vectors resulted in a very low estimate of B_{MSY}/K^{sp} of 0.10 for *M. paradoxus*, but it was found that a simple increase of the basal mortality rate for older fish from 0.20 to 0.25 was sufficient to increase this B_{MSY}/K^{sp} estimate to 0.17. This document provides the details of the mortality-at-age vectors that are being used as the base case assumption for the OMs being for testing OMP-2022. It shows the impact of using these vectors in the RC model, both on the assessment results and on OMP-2018 projections. The updated RC has an improved negative log-likelihood score, higher absolute values for the *M. paradoxus* spawning biomass and slightly lower estimates of B^{sp}/B_{MSY} trajectories for *M. paradoxus*, although this ratio is still well above one at present. The projections of the two models under use of the OMP-2018 rules to set future TACs are not very different, apart from the Update with the new mortality-at-age vectors resulting in slightly wider probability envelopes. An earlier version of this document was presented to the hake Task Team, and as there was no obvious reason to reject the proposed new BC mortality-at-age vectors, these have been accepted for use for the conditioning of the Reference Set models and OMP-2022 development. Sensitivity to alternative mortality-at-age vectors will be considered in robustness tests.

Keywords: Hake, OMP-2022, mortality-at-age, predation model

Introduction

The mortality-at-age vectors which have been used in the hake single-species model since 2018 assume an age-dependent basal mortality vector which decreases linearly from 0.7 at age 0 to 0.2 at age 5 and stays at 0.2 thereafter[2]. The predation mortality rates estimated in the predation model were added to this basal mortality rate, and the total natural mortality-at-age vectors averaged from 1984 onwards have been used in the single-species model since then. The vectors were also smoothed, such that each value was replaced by the average of that value and the values immediately before and after (i.e. average of three values, except for the first and last values which are the average of only two values respectively).

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[2] The reason for the increased basal mortality rate at smaller ages was to try to achieve a monotonically decreasing total natural mortality-at-age vector (Ross-Gillespie and Butterworth, 2018).

Since 2018 several updates have been made to the hake predation model, including use of the updated growth curves which assume that the growth rate is halved from age 9 onwards (Ross-Gillespie and Butterworth, 2019). For the OMP 2022 revision, the corresponding updated mortality-at-age vectors should replace those used previously.

However, applying the straightforwardly updated mortality-at-age vectors (i.e. assuming the same basal mortality vector as before with $M_0 = 0.7$ decreasing linearly to 0.20) to the single species Reference Case (modified Ricker) model resulted in a very low estimate of B_{MSY}/K^{sp} of 0.10 for *M. paradoxus* (in contrast to the value of 0.16 obtained with the original 2018 mortality-at-age vectors). In the past, an OM with such a low estimate for this quantity would have been excluded from the Reference Set.

The values of the basal mortality rate vector components are one of the major assumptions made in the hake predation model, as they cannot be based on data or observations, so instead are based rather on what would generally be considered reasonable. Several adjustments to the basal mortality rate vector were explored to try to find alternative vectors that result in a higher estimate of B_{MSY}/K^{sp}. Possible alternative basal mortality vectors included ones with the same structure as the 2018 vector, but with different values for M0 and M5, age-independent vectors (i.e. a single value for all ages), and even age- and species-dependent vectors where different assumptions where made for each species. In the end, it was found that a simple increase of the M5 value from 0.20 to 0.25 was sufficient to increase the B_{MSY}/K^{sp} estimate for *M. paradoxus* from 0.10 to 0.17. In the interest of maintaining as much continuity as possible, this vector was proposed to and accepted by the Task Team for the Base Case (BC). Other choices for mortality vectors may be considered in robustness tests for the OMP-2022 revision.

The 2021 Reference Case (RC) model from Ross-Gillespie and Butterworth (2021) was re-fit using the proposed new BC mortality-at-age vectors. Key results have been included in this document for the two runs: (1) the RC model with the original 2018 mortality at age vectors (as presented in Ross-Gillespie and Butterworth, 2021), and (2) the same model but using the updated mortality-at-age vectors. Both models have also been projected forward under TACs set by the OMP-2018 rules to assess the impact of updating the mortality-at-age vector on these projections.

Note that since 2018 the single-species hake assessment models have been using the current two-line form for the growth curves which assume slower growth for older hake. The 2018 hake predation model used the original straight line growth curves, but the updated predation model also uses the current two-line growth curves with slower growth for older hake.

Results

Table 1 lists key assessment outputs for the 2021 RC model and the update using the proposed new BC mortality-at-age vectors. Table 2 lists the negative log-likelihood components for the two models.

Figure 1 shows the mortality-at-age vectors being proposed as the BC for the 2022 OMP revision, along with alternative vectors $M_5 = 0.2$ as was assumed previously but leads to an *M. paradoxus* estimate for B_{MSY}/K^{sp} of 0.10. The 2018 mortality-at-age vectors are also shown.

Figure 2 - Figure 8 show various assessment outputs for the two model fits.

Figure 9 shows the projections of the two models under use of the OMP-2018 rules to set future TACs. Note that these projections are for the original OMP-2018 rules with no adjustment to the *b* parameters and the normalisation approach as used for OMP-2018.

Discussion

Some key observations are listed in the bullet points below.

- A key difference between the 2018 vectors and the proposed BC is that the 2018 *M. paradoxus* vector had a higher mortality for the middle ages (4 to 8) and lower mortality at the lower ages. While the mortality rate for the lower ages will have less of an impact on the hake stock assessment (recruitment is predominantly driven by the larger fish through the spawner biomass), the lower mortality rate for the middle ages that is estimated now is likely responsible for the lower B_{MSY}/K^{sp} estimate and the reason for the M_5 value needing to be increased. The difference in the shape of these mortality-at-age curves is most likely a result of the updated growth rates – the new curves assume slower growth for older fish, which in turn will impact the age of the preferred prey fish assumed by the model and the overall estimated mortality rates.
- Changing the mortality-at-age vectors is a fundamental change to the model, so some non-trivial changes to the assessment output is to be expected. Some notable changes include the following.
    - A higher absolute value of spawning biomass is estimated for *M. paradoxus* with the new mortality-at-age vectors, although the trajectories relative to pristine biomass are very similar (Figure 2).
    - The new estimate of B^{sp}/B_{MSY} for *M. paradoxus* is lower, although still well over 1 (1.47 compared to the 2021 RC model estimate of 1.80, see Figure 2).
    - The negative log-likelihood is improved when using the new mortality-at-age vectors (Table 2). There is a notable improvement to the South Coast ICSEAF CPUE fit (Figure 5).
- The projections of the two models under the OMP-2018 rules are not very different, including for future TACs, aside from the Update with the new mortality-at-age vectors resulting in slightly wider probability envelopes.
- As there was no obvious reason to reject the proposed new BC mortality-at-age vectors, the Task Team accepted this as the new BC going forward. Sensitivity to alternative mortality-at-age vectors will be considered during the OMP2022 revision process.

References

Ross-Gillespie, A. and Butterworth, D.S. 2018. Update on the MARAM hake predation model, focusing on the natural mortality-at-age vectors to be used for the Reference Set of Operating Models for the 2018 Hake OMP review. DEFF Fisheries document: FISHERIES/2019/MAR/SWG-DEM/11: 13pp.

Ross-Gillespie, A. and Butterworth, D.S. 2019. Updated specifications, conditioning results and projections fro the Hake OMP2018 Reference Set models. DEFF Fisheries document: FISHERIES/2019/MAR/SWG-DEM/03: 49pp.

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Table 1: Key assessment outputs for the 2021 RC model from Ross-Gillespie and Butterworth (2021) and from the same model using the updated mortality-at-age vectors being proposed for the BC going forward. Biomass units are thousand mt.

M. paradoxus	(1) 2021 RC	(2) Update	*M. capensis*	(1) 2021 RC	(2) Update
K^{sp}	338	535	K^{sp}	346	299
B_{MSY}	55	93	B_{MSY}	96	94
MSY	139	131	MSY	79	80
B^{sp}_{2021}	91	138	B^{sp}_{2021}	255	234
B^{sp}_{2021}/K^{sp}	0.27	0.26	B^{sp}_{2021}/K^{sp}	0.74	0.78
B^{sp}_{2021}/B_{MSY}	1.67	1.47	B^{sp}_{2021}/B_{MSY}	2.67	2.49
B_{MSY}/K^{sp}	0.16	0.17	B_{MSY}/K^{sp}	0.28	0.31

Table 2a: Negative log-likelihood values for the RC model (1) using the original 2018 mortality-at-age vectors and (2) the updated mortality-at-age vectors.

Run	GLM CPUE	ICSEAF CPUE	Survey abun.	Comm. CAL	Survey CAL	Recruit. resid.	ALKs	Penalties	Total (w/o pen.)
(1) October 2021 RC	-226.73	-37.36	-38.65	-1570.34	-1649.08	9.05	130.26	0.09	-3382.84
(2) Update	-227.81	-40.07	-41.05	-1572.48	-1649.25	9.68	129.52	0.02	-3391.47

Table 2b: The first row of this Table lists the negative log-likelihood for the October 2021 RC model (1), while the second row shows the difference in negative log-likelihood values between the updated run (2) and the original (1), with a negative value indicating a better negative log-likelihood for the update.

Run	GLM CPUE	ICSEAF CPUE	Survey abun.	Comm. CAL	Survey CAL	Recruit. resid.	ALKs	Penalties	Total (w/o pen.)
(1) October 2021 RC	-226.73	-37.36	-38.65	-1570.34	-1649.08	9.05	130.26	0.09	-3382.84
(2) Update	-1.08	-2.71	-2.40	-2.14	-0.18	0.62	-0.74	-0.07	-8.63

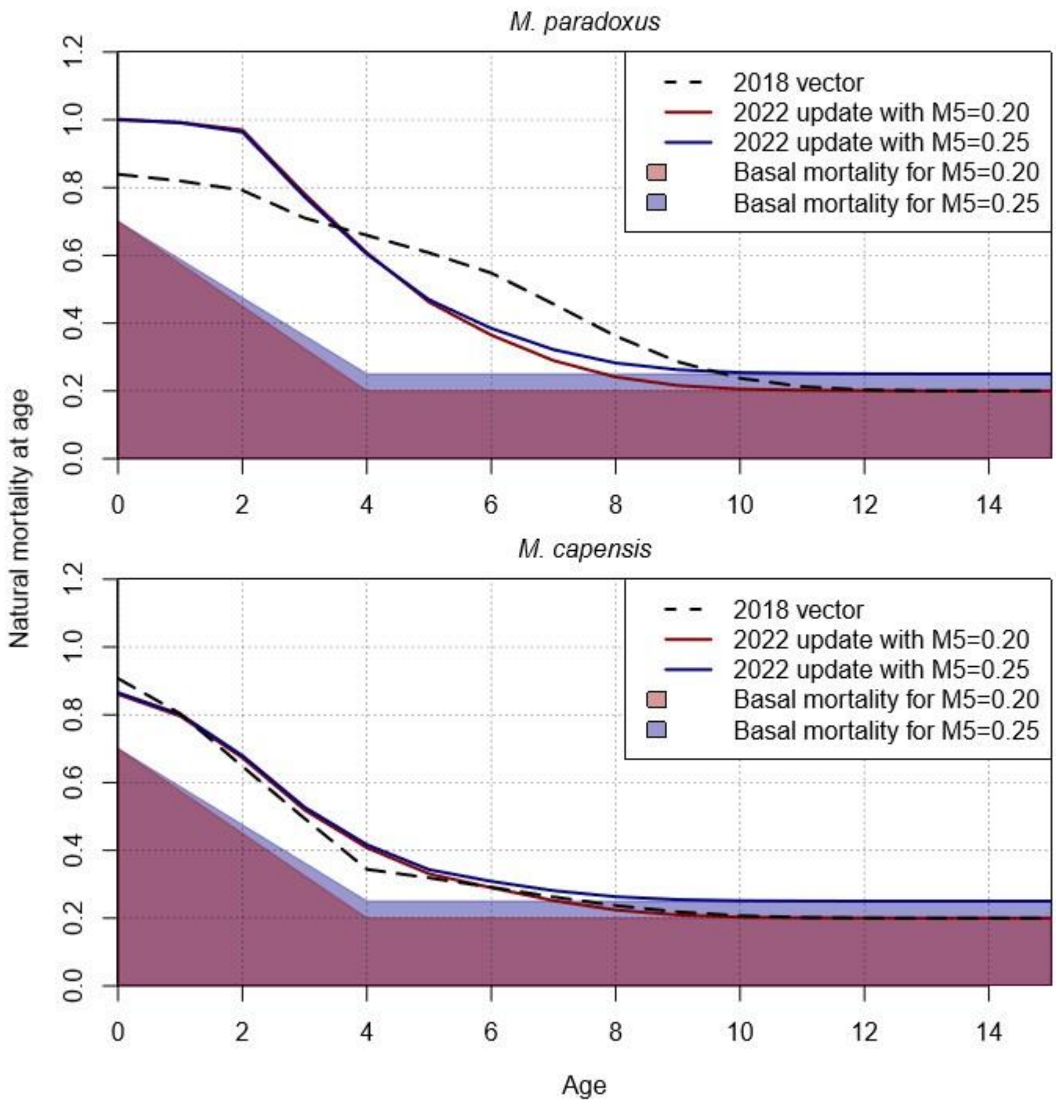

Figure 1: The total natural mortality-at-age vectors being proposed for the 2022 OMP revision are shown in blue. These vectors correspond to a basal mortality rate decreasing linearly from a value of 0.7 at age 0 to 0.25 at age 5. The red curve shows the mortality-at-age vectors estimated when the M_5 value is 0.2 (as was assumed for the 2018 mortality-at-age vectors, which are shown by the black dashed lines).

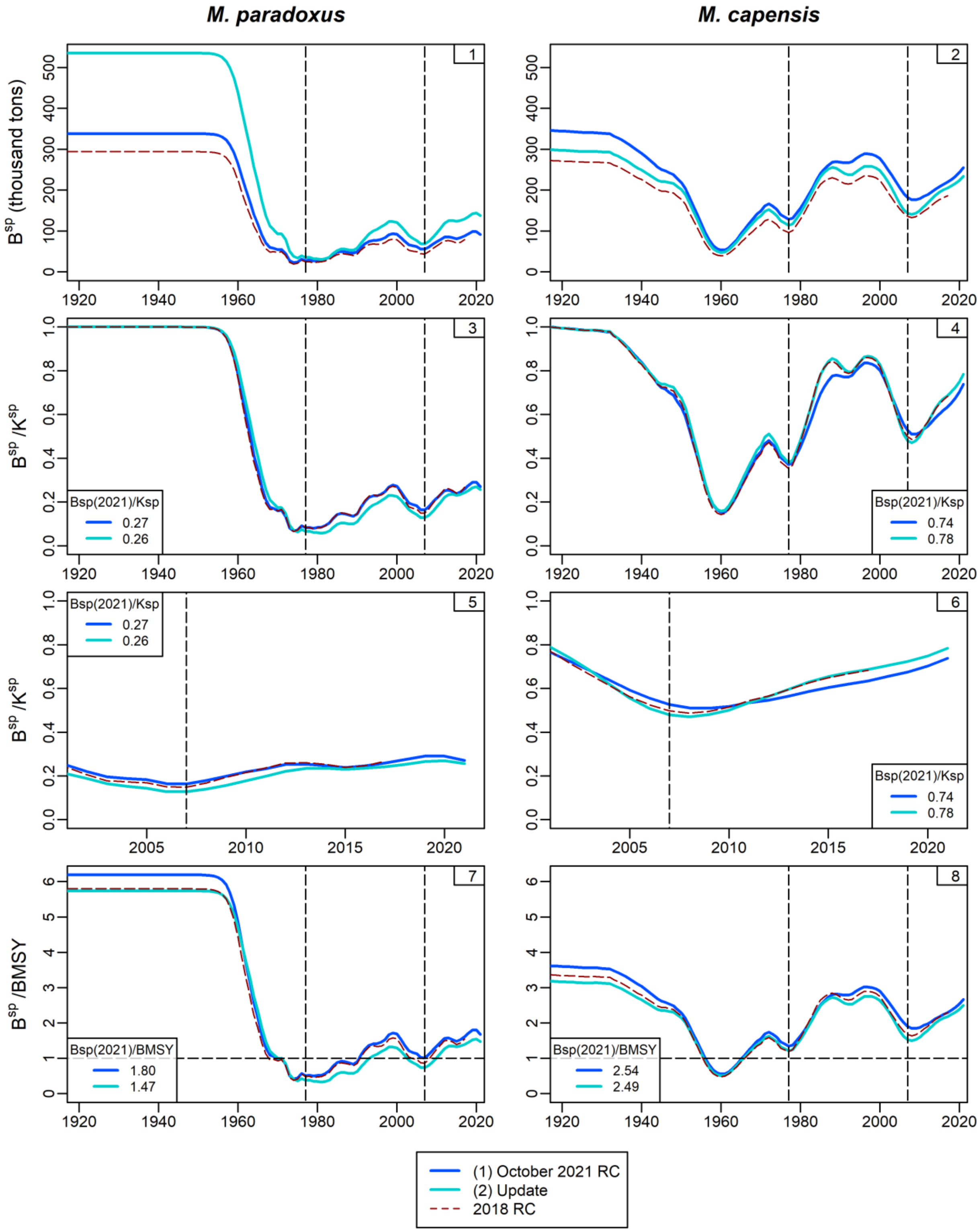

Figure 2: Female spawning biomass is shown for each species in absolute terms (top row), relative to pristine spawning biomass (second row), relative to pristine spawning biomass but for the 2000-2021 time period only (third row) and relative to B_{MSY} (fourth row). The 2018 RC trajectories from Ross-Gillespie and Butterworth (2019) have been included for comparison purposes; they are shown by the dashed red lines.

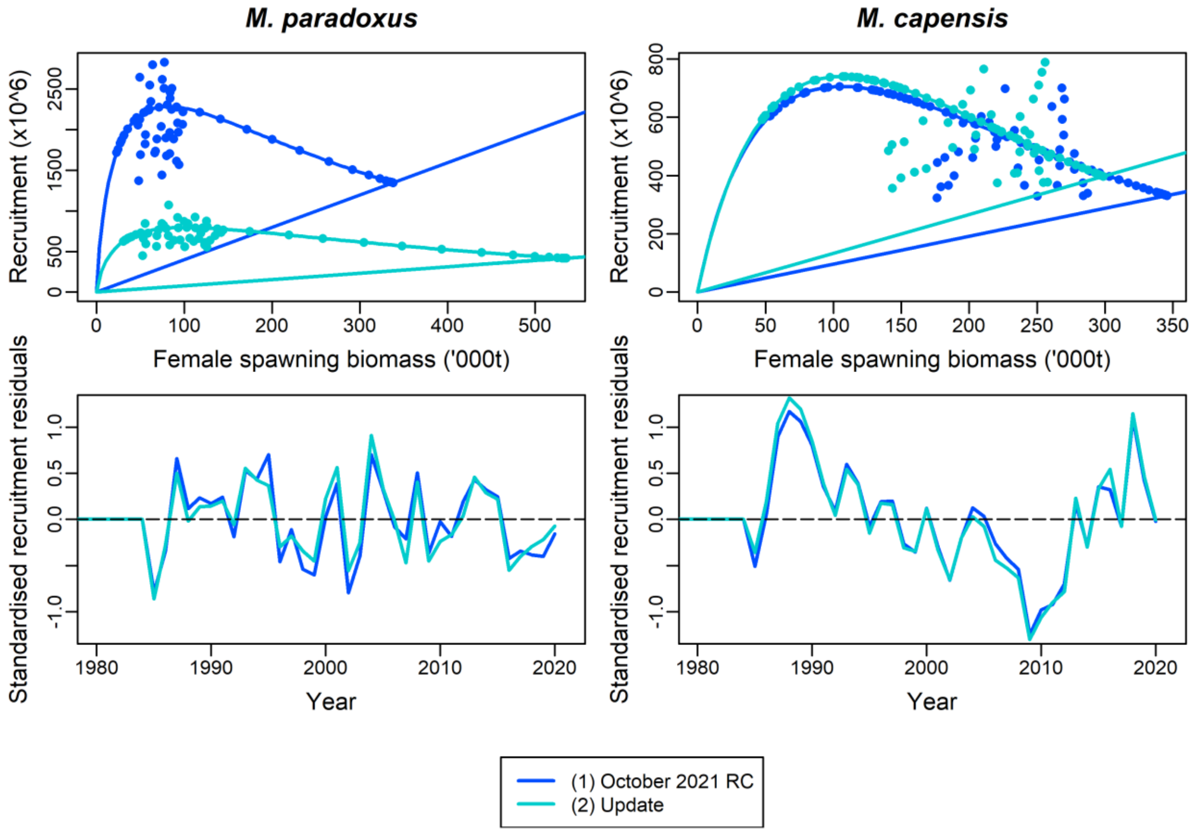

Figure 3: The top row shows the estimated recruitment-spawning biomass relationship for the two assessment models, while the bottom row shows the standardized estimated recruitment residuals.

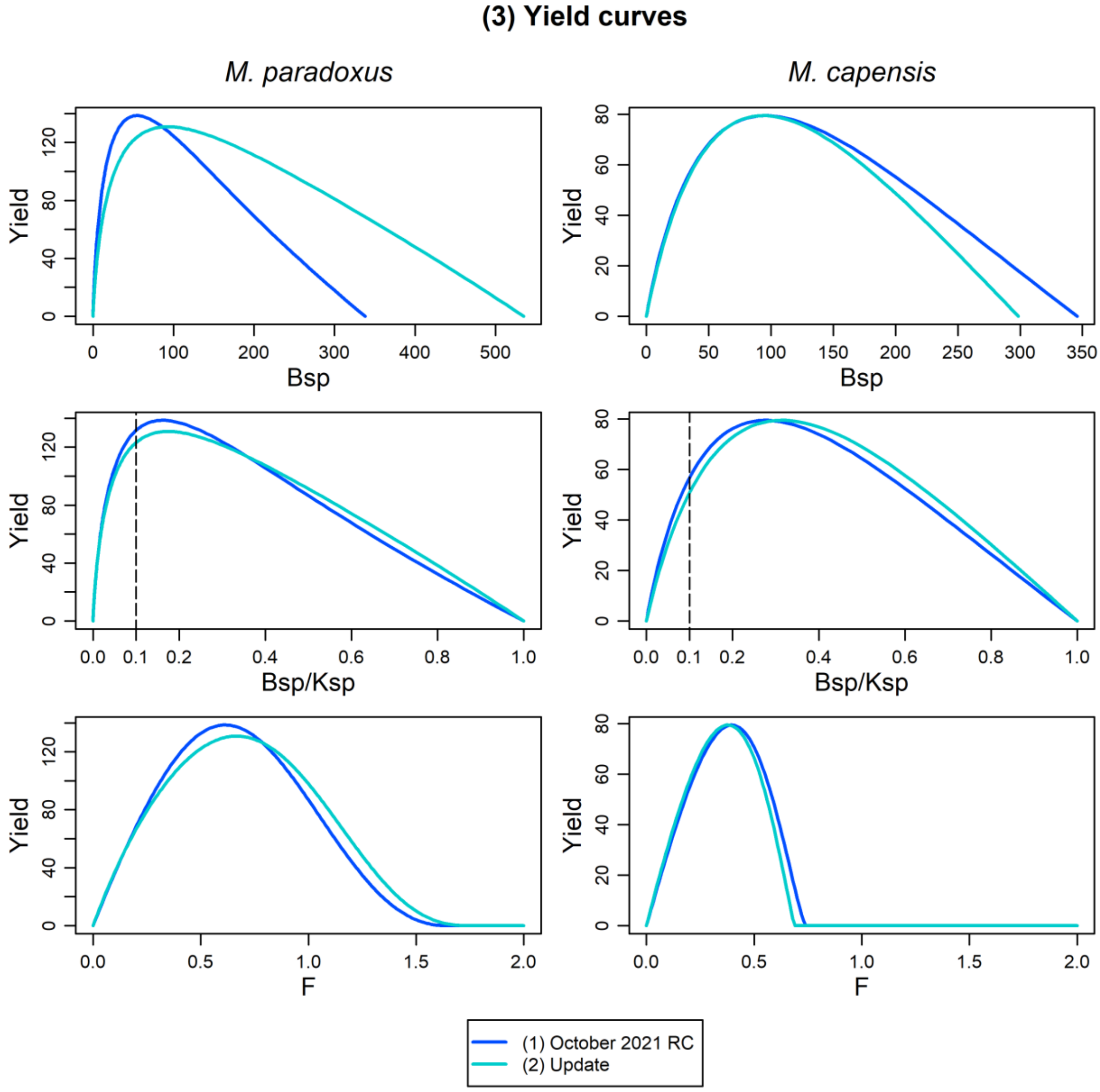

Figure 4: The estimated yield curve (yield in thousand tons) is plotted against female spawning biomass (top panel), female spawning biomass relative to pristine levels (middle panel) and fishing mortality rate F (bottom panel).

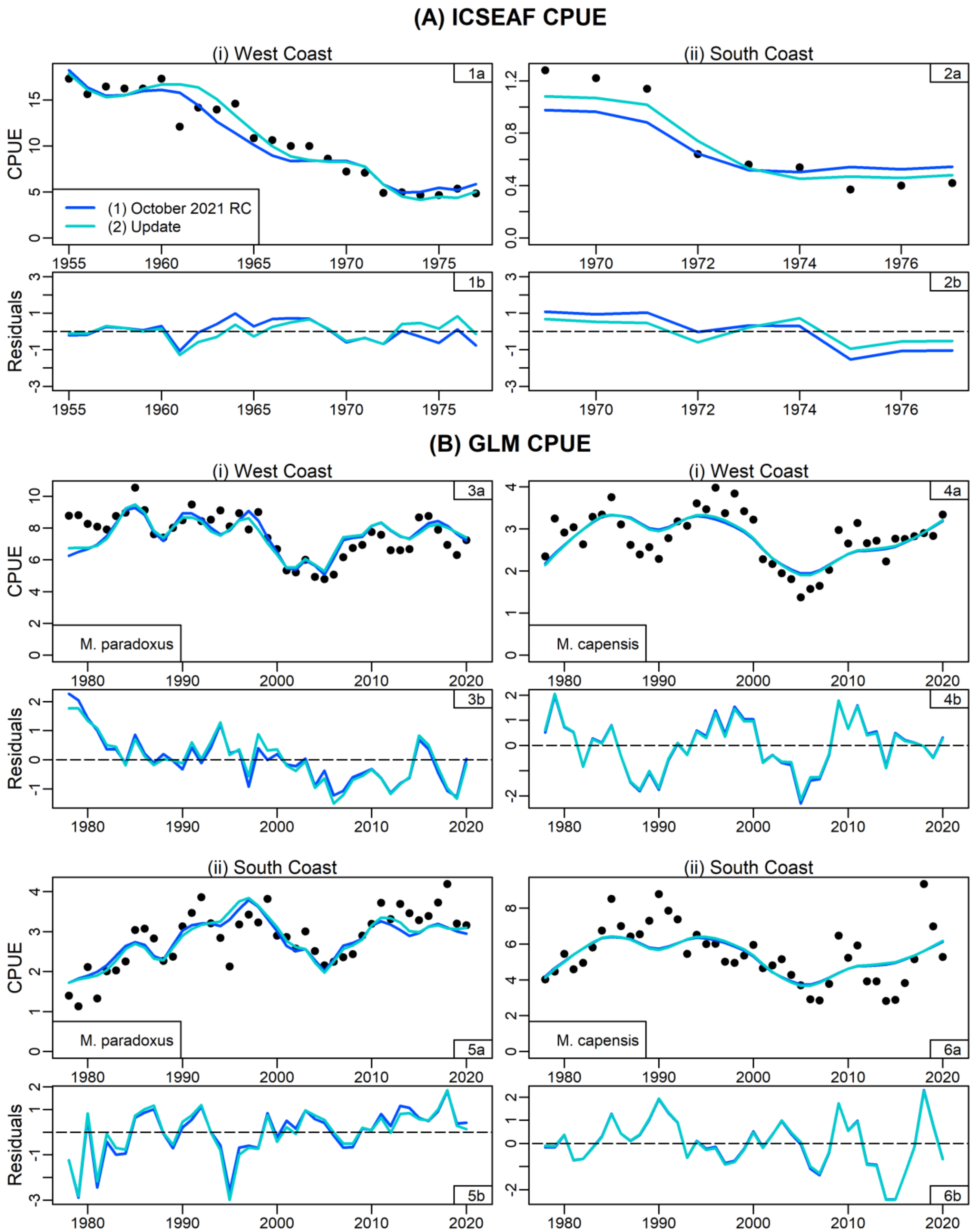

Figure 5: Fits to (A) the historical ICSEAF CPUE data and (B) the commercial GLM-standardized CPUE data are shown. Standardised residuals are shown underneath each plot.

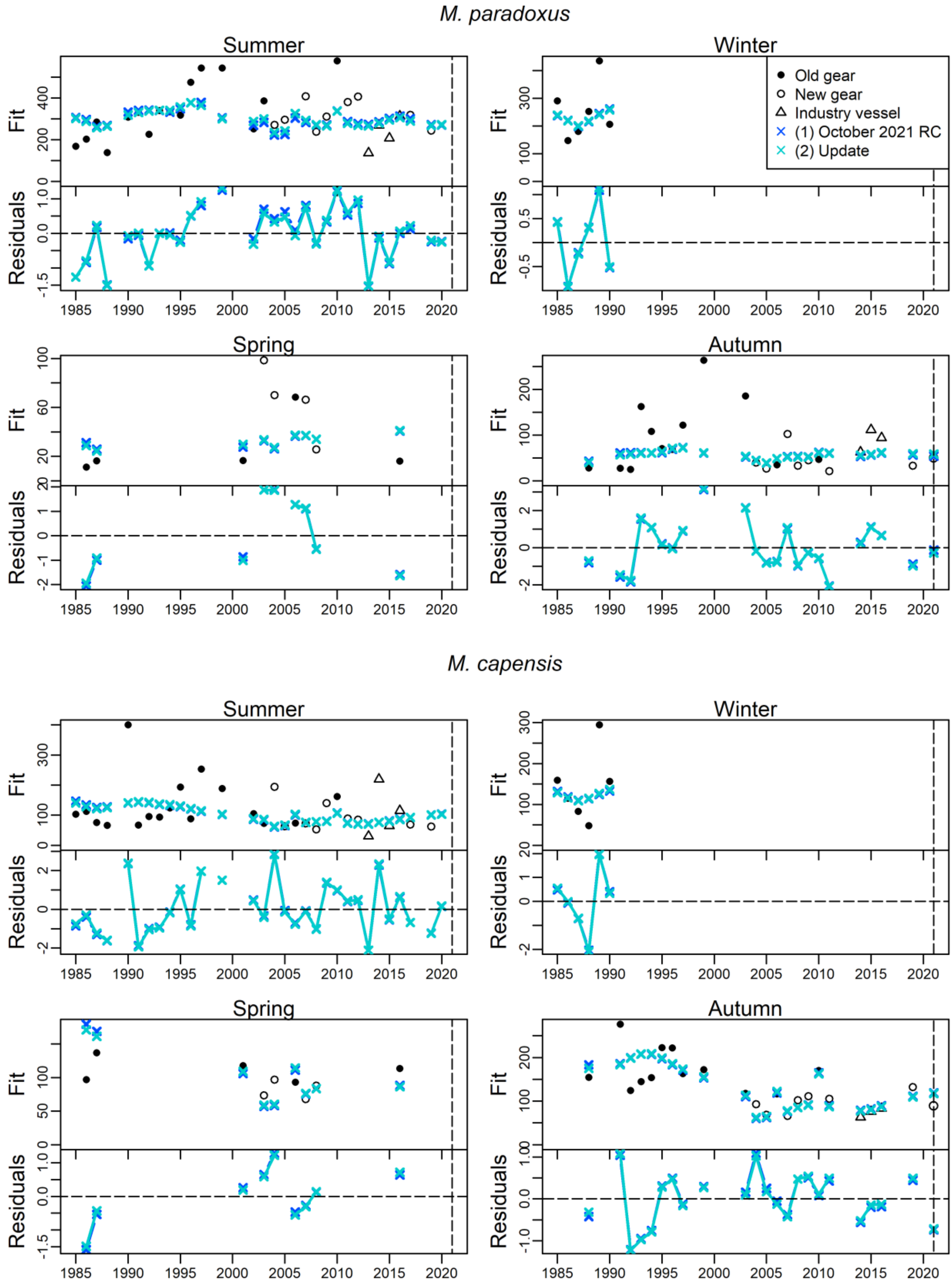

Figure 6: Fits to the survey relative abundance series are shown for the 2021 RC assessment model and its update. Standardised residuals are shown underneath each plot. Note that at this level of resolution, the two fits can scarcely be distinguished.

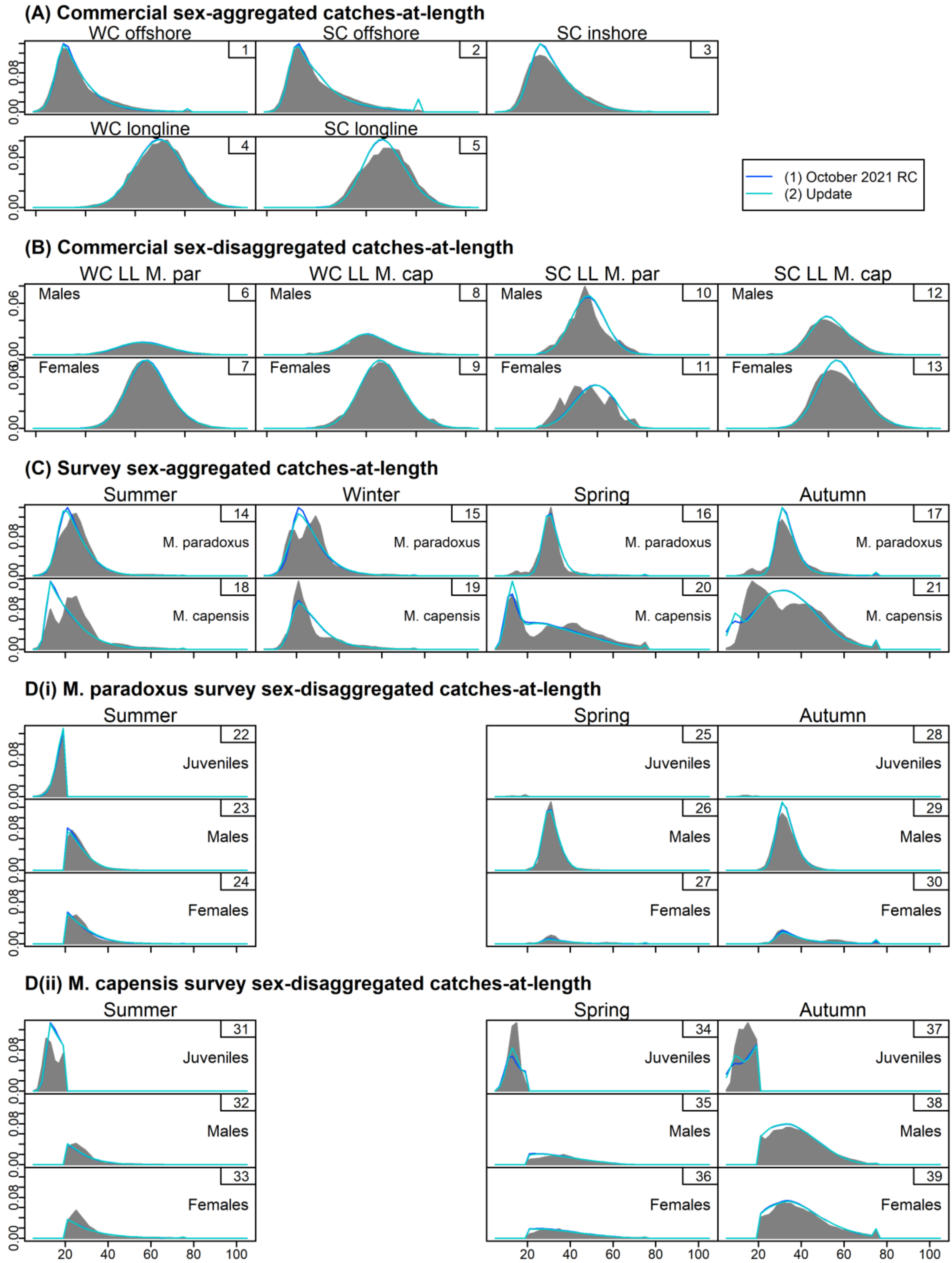

Figure 7: Fits to commercial and survey commercial catch-at-length data are shown, with the grey shaded area showing the distributions of the data and the blue lines the model fits. The data and model fits have been averaged over the years for which data are available. Note that at this level of resolution, the two fits can scarcely be distinguished.

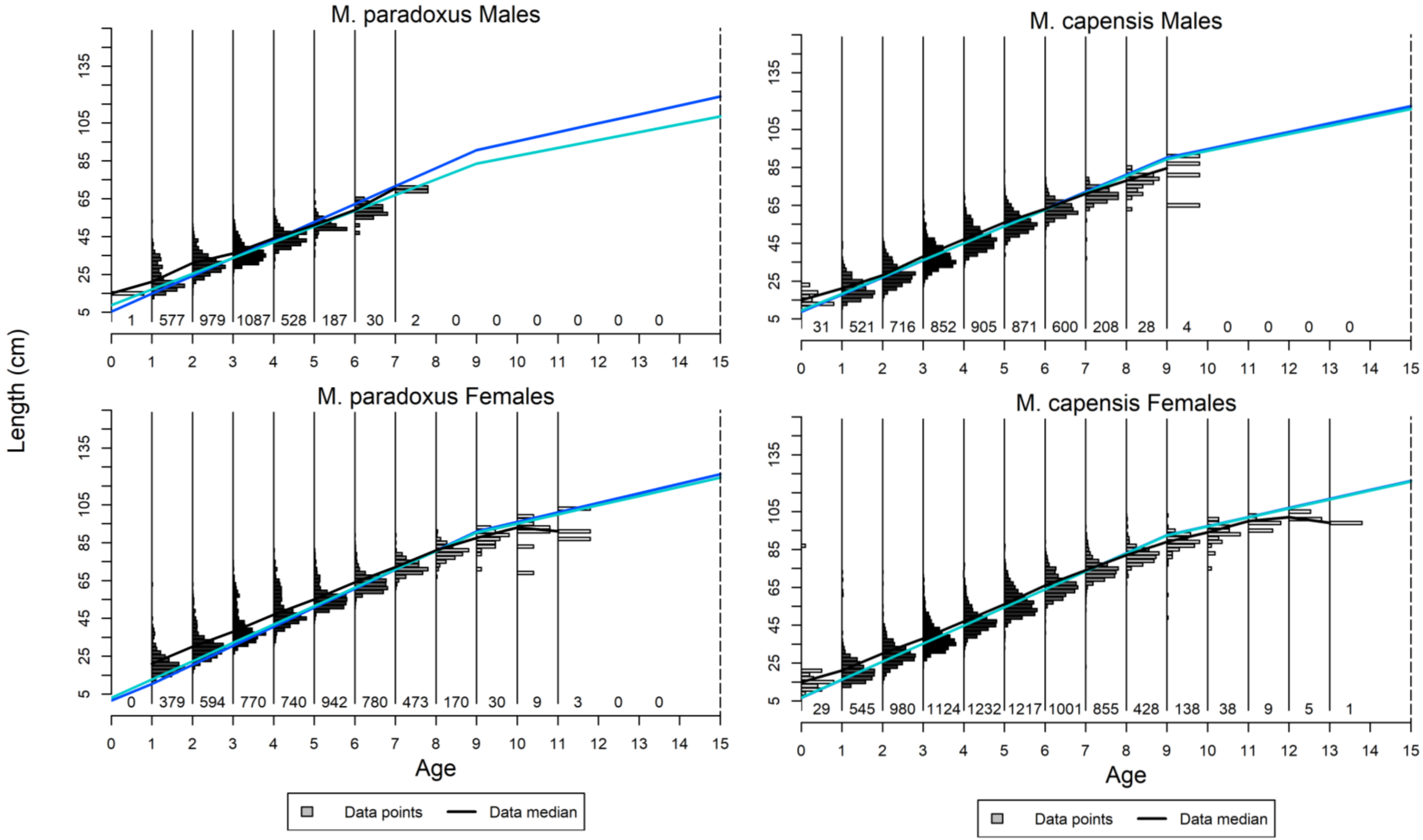

Figure 8: The estimated growth curves are superimposed on the length-at-age distributions derived from the age-length-key data. The numbers printed at the bottom of each age class indicated the number of data points available, and the shading of the bars is relative to these numbers, with darker shades indicating more data points. As for previous Figures, the dark blue lines are for the October 2021 RC with the original 2018 mortality-at-age vectors and the light blue lines for the same model using the updated mortality-at-age vectors.

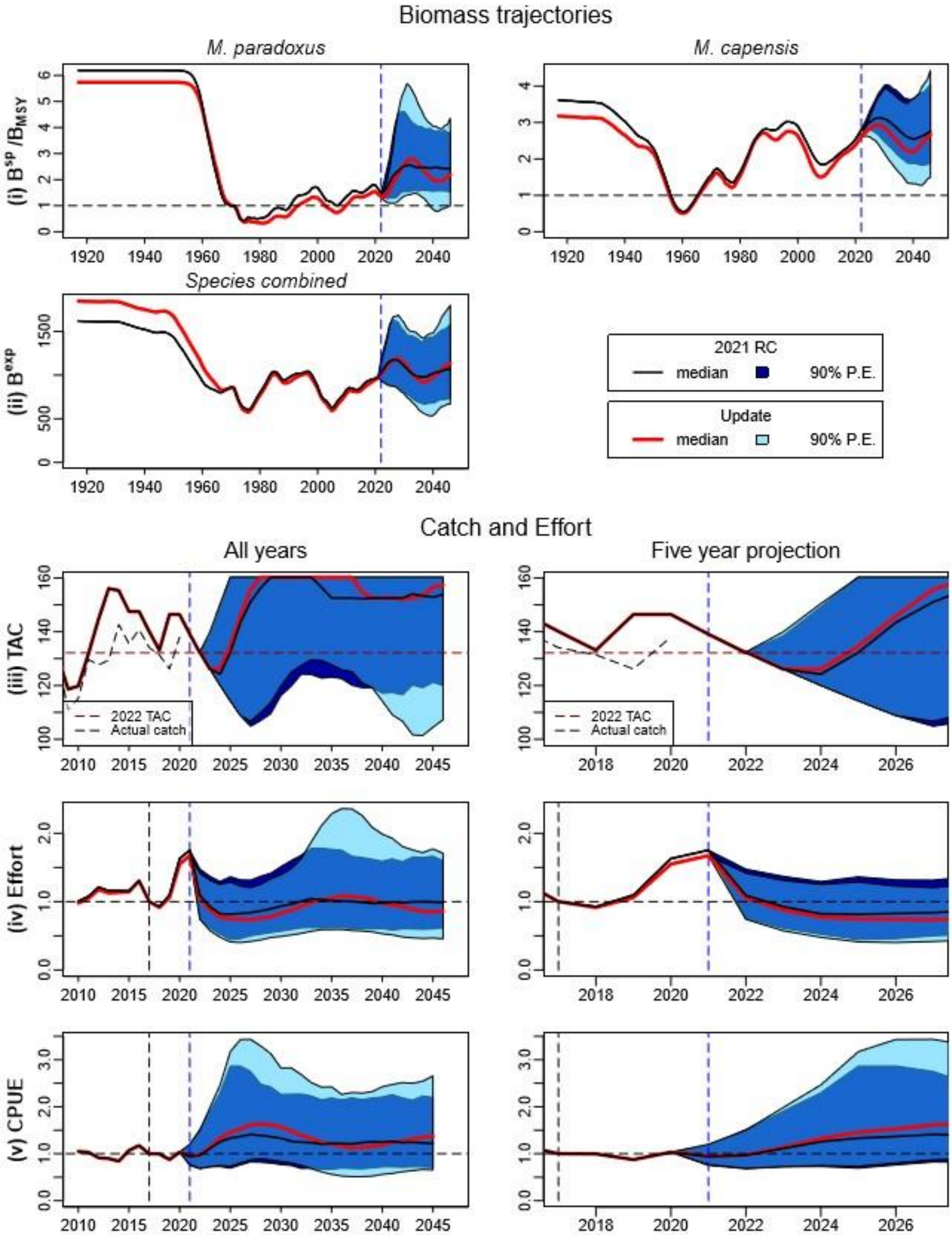

Figure 9: The 2021 RC and the update with the new mortality-at-age vectors are projected forward using the OMP-2018 rules to set future TACs. The top row shows the spawning biomass relative to B_{MSY} for each species, with the species-combined exploitable biomass in the second row. The next three rows show the projected TAC, effort and CPUE, with the left column showing all the projected years and the right column the next five years. Results for the 2021 RC are shown by black lines (medians) and dark blue shaded areas (90% probability envelope) and the results for the update are shown by the red lines and the light blue shaded areas. Note that for the TAC plots, the actual past catches have been shown by the dashed black lines, as these differ at times from the TAC recommendations.